

Avery
Briggs

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Dear Outlier AI Team,

I am writing to express my interest in the Full Stack Software Engineer Expert role for AI Training. The opportunity to evaluate code quality, analyze software engineering practices, and contribute to improving AI-generated outputs strongly aligns with my professional experience and interests.

I hold a Bachelor of Computer Science from the University of New Brunswick and have over three years of experience building and maintaining production-grade software systems in a manufacturing environment. In my role as a Software Systems Developer and Process Analyst at BWS Manufacturing, I developed full-stack internal tools using Python, SQL Server, and modern UI frameworks such as Streamlit. These tools supported operational workflows across procurement, production, inventory, and sales.

A significant portion of my work involved debugging complex issues, analyzing system behavior, and ensuring data integrity across interconnected systems. I regularly evaluated code quality, refactored legacy applications, and implemented structured solutions to improve maintainability and performance. This included consolidating multiple Access-based systems, developing OCR pipelines for unstructured data extraction, and designing interactive dashboards for operational analysis.

I am particularly drawn to this role's focus on reviewing and improving AI-generated code. My experience working across multiple technologies—including Python, SQL, VBA, and Java—has given me a strong foundation in identifying edge cases, enforcing best practices, and communicating technical feedback clearly and effectively. I am comfortable working independently with detailed guidelines and enjoy applying structured reasoning to technical evaluation tasks.

This opportunity represents a compelling intersection of software engineering and emerging AI systems, and I would welcome the chance to contribute to improving model quality through expert analysis and feedback.

Thank you for your time and consideration.

Sincerely,
Avery Briggs

Professional Summary

Full-stack software developer with 3+ years of experience building and maintaining production-grade systems across data, backend logic, and user-facing tools. Strong background in code quality analysis, debugging, and system design, with hands-on experience evaluating complex workflows and improving maintainability of legacy systems. Experienced in evaluating code quality, debugging complex systems, and communicating technical reasoning clearly.

Experience

Software Systems Developer | Process Analyst

BWS Manufacturing Ltd. | Centreville, N.B. (Jan 2021 – Feb 2026)

Developed internal software systems and software development tools supporting operational workflows across Sales, Procurement, Production, and Scheduling within a small technical team in a manufacturing environment.

Key Contributions

- Designed and developed full-stack internal tools supporting production, procurement, inventory, and sales workflows using Python, SQL Server, and Streamlit.
- Analyzed and debugged complex system behaviors across interconnected databases and applications, identifying root causes and implementing reliable fixes.
- Evaluated and refactored legacy Microsoft Access/VBA applications, consolidating multiple systems into a maintainable architecture and improving code reuse.
- Developed structured data processing pipelines, including OCR-based extraction systems for converting unstructured PDF data into usable formats.
- Built interactive dashboards and analytical tools to support decision-making, requiring careful validation of logic, edge cases, and data integrity.
- Wrote and optimized SQL queries for large datasets, ensuring performance and correctness across production systems.
- Provided technical support and troubleshooting across software, ERP integrations, and infrastructure, often diagnosing issues without prior documentation.
- Collaborated across departments to translate business requirements into reliable software solutions, emphasizing clarity, maintainability, and usability.
- Reviewed, tested, and validated code logic across multiple internal tools, ensuring correctness, handling edge cases, and improving reliability in production environments.

Notable Projects:

- IT Request and Inventory ticketing system. Built using SQL and MS Access.
- Developed an Access macro and VBA workflow to automate batch report generation, triggered from a Python/Streamlit dashboard. Implemented timestamp validation and fault-tolerant logic to ensure reliable generation of up to 50 reports per execution.
- Yellow Tag system to automatically flag, record, and report missed parts when picked for production. Offers a variance reconciliation opportunity earlier in the production pipeline, highlighting common trends. Front-end built using an automatically refreshing form presenting live Syspro Data.
- Built OCR-based data extraction pipelines using Python (Pytesseract, Pdfplumber) to transform legacy documents into structured datasets.
- Developed a warehouse visualization system using Plotly to model spatial inventory layouts and improve operational decision-making.

Other Job Experience:

- Basketball Referee – CBOC – Fredericton, NB – Oct 2013 – Mar 2020 – Part-time.

Education

Bachelor of Computer Science

UNB – Fredericton N.B. (Sept 2017 – May 2020)

- Specialization: Software Systems.
- Recipient of Governor Thomas Carleton Entry Scholarship.

Skills

- **Languages:** English (native), French (Intermediate professional proficiency).
- **Programming:** Python, VBA, SQL, Java.
- **Databases:** Microsoft Access, Microsoft SQL Server.
- **Software Engineering:** Debugging, code review, refactoring, system design, data validation.
- **Frameworks / Libraries:** Streamlit, Pandas, Plotly, CustomTkinter, Pdfplumber, Pytesseract.
- **Learning:** TypeScript, C#, .NET, other modern web frameworks.
- **Tools:** Microsoft Excel, Git, Syspro ERP, ShopClock.

Awards & Activities

- Finalist (Top 12) - Kaspersky Academy Partner's Summit (KAPS) Lisbon, Portugal (December 2019).
- Certified intermediate plus in the French oral proficiency exam (June 2013).
- UNB dormitory Orientation Committee (September 2014) & Social Representative (September 2014 – April 2015).
- UNB Relay for Life steering committee logistics team (September 2013 – February 2014).

Hobbies

3D printing and hardware projects, sports analytics software, recreational athletics (cycling, kayaking, basketball officiating).

Projects

- NHL data analysis dashboard displaying real-time game data and standings using the NHL Stats API.
- STL file generation for automating the creation of strings of text using custom fonts and other parameters. Uses a Solid2 to generate files, and Streamlit interface to model and tweak them.
- Minesweeper Android mobile application, built using Java and Android Studio.
- Image editing Android application taking advantage of the Gestures library to offer a unique interface for tuning photos.
- Traffic simulation engine using Pygame. Mimics some elements from Cities Skylines traffic patterns.
- COVID-19 interaction simulator built with JavaFX modeling contagion spread in configurable populations.

References available upon request